Types of Neural Networks, History and Applications of Deep Learning

Contents

1. Overview . . . 2
2. Types of neural networks . . . 2
   - 2.1 Multi-layer perceptron (MLP) . . . 3
   - 2.2 Convolutional neural network (CNN) . . . 4
   - 2.3 Recurrent neural network (RNN) and LSTM . . . 4
   - 2.4 Autoencoder . . . 5
   - 2.5 Generative adversarial network (GAN) . . . 5
3. History of deep learning . . . 6
   - 3.1 The perceptron and the first AI winter . . . 6
   - 3.2 Backpropagation and the second boom . . . 7
   - 3.3 Universal approximation . . . 8
   - 3.4 The second AI winter . . . 9
   - 3.5 Deep belief networks and the name "deep learning" . . . 9
   - 3.6 2012: ImageNet and after . . . 9
4. Applications of deep learning . . . 11
   - 4.1 Mainstream applications . . . 11
   - 4.2 Image, sound and text applications . . . 13
   - 4.3 Generative applications . . . 13
   - 4.4 Everyday industry uses . . . 13
5. Summary . . . 14
6. Sources . . . 15
7. Key terms . . . 16

Where this fits: Pipeline map steps 0 (Foundations), 8 (Model). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	**8 Model**	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** What deep learning is.
- **Leads to:** Multi-layer perceptron (MLP).

1. Overview

Key point: Five main types of network (MLP, CNN, RNN, **autoencoder** (G-231), GAN) each suit a kind of data or task. Neural networks rose and fell twice before deep learning took off in 2012, and today they power cars, games, assistants and creative tools.

This Note gives that big picture before the technical Notes start. The Note has three parts:

1. **Types of neural networks:** what each of the five main types is for (Figure 1).
2. **History:** from the first **perceptron** (G-1486) in 1958 to the ImageNet breakthrough of 2012 (Figure 3).
3. **Applications:** where **deep learning** (G-568) is used today.

Deep learning has parts of a network (input, hidden and output layers) and five forces behind its success, both explained earlier. This Note builds on both.

2. Types of neural networks

Key point: MLPs for general supervised learning, CNNs for images, RNNs and **LSTMs** (G-1123) for sequences, autoencoders for compression, GANs for generating new data.

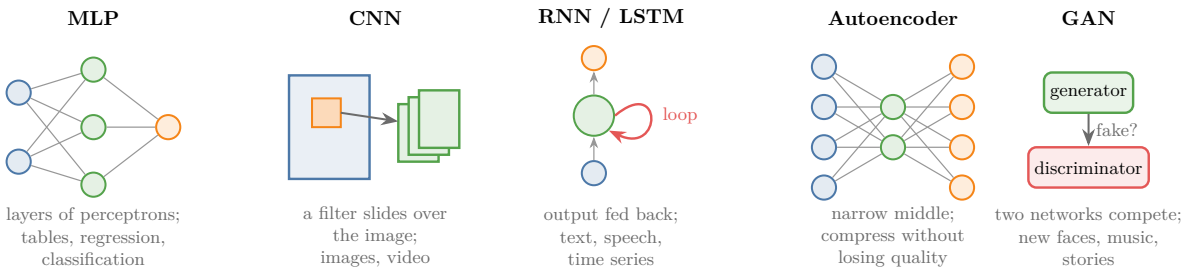


Figure 1: The five main types of neural network

There are many types of neural networks, and each type is a large subject on its own. Figure 1 shows the structure of the five main ones.

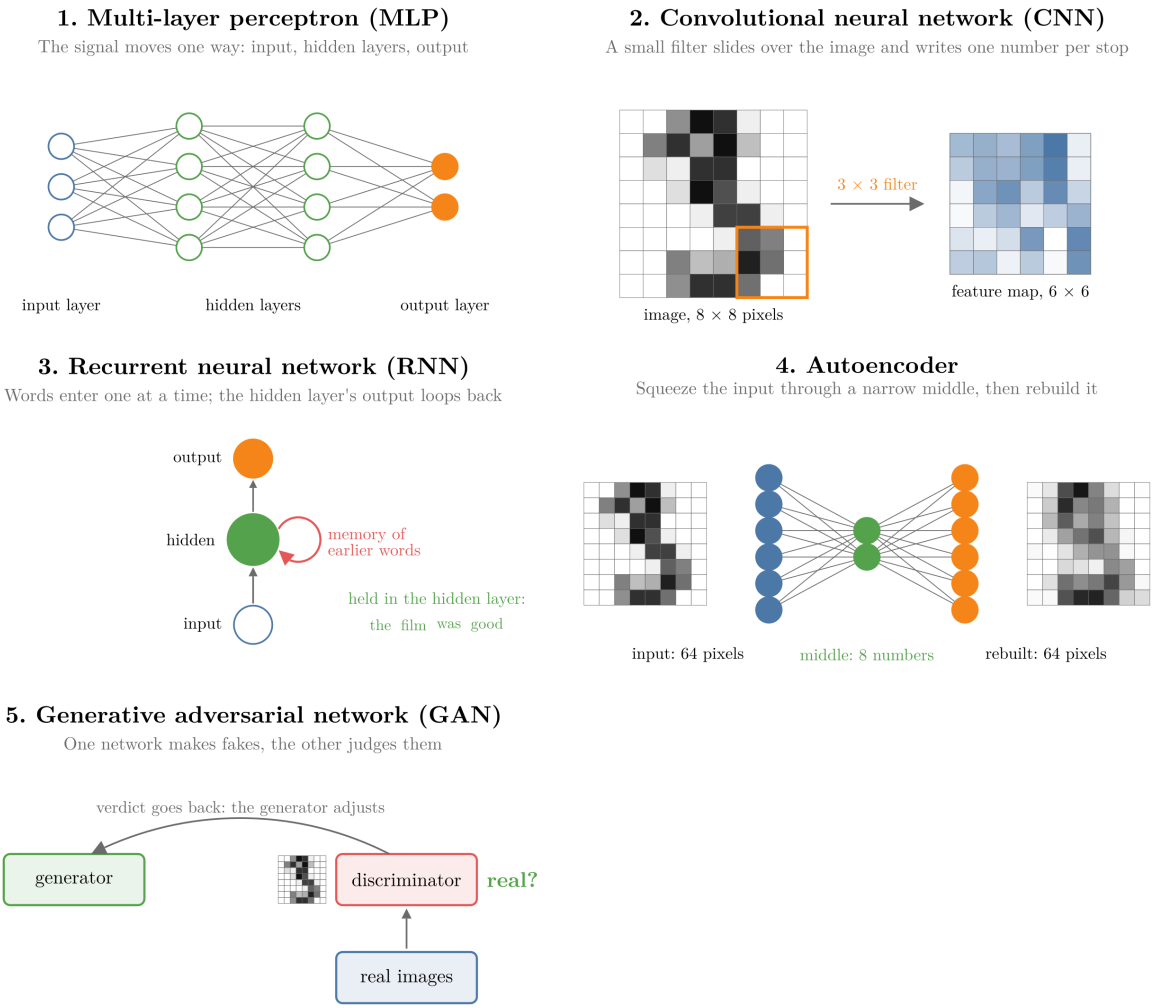


Figure 2: How data moves through each of the five types, one type at a time. The digit is a real 8 × 8 image; the filter values and the rebuilt digit are computed, and the GAN part is a schematic

Figure 2 animates the same five types. In each part, watch where the orange signal travels: that path is what makes the types differ. The subsections below take the parts one by one.

2.1 Multi-layer perceptron (MLP)

> **Key point: Perceptrons** (G-1486) organised in layers; the simplest network, good at non-linear relationships in ordinary supervised problems.

The simplest network stacks small units in layers and passes the data straight through them. In part 1 of Figure 2, the signal starts at the input layer, crosses the two hidden layers and ends at the output layer, always in one direction.

The units are perceptrons, so the network is called a **multi-layer perceptron (MLP)**

(G-1270): many perceptrons organised in an input layer, one or more hidden layers and an output layer. Until the other types appeared, "MLP" and "artificial neural network" meant the same thing.

MLPs work on any supervised problem (learning from examples with known answers), regression (predicting a number) or classification (predicting a class). Adding hidden layers helps them capture non-linear relationships, which a single perceptron cannot (combining two perceptrons shows why).

2.2 Convolutional neural network (CNN)

> **Key point:** At least one convolutional layer; the standard network for images and video.

An image is a grid of pixels, and what matters in it (an edge, a corner) is small and can sit anywhere. So the network looks at the image through a small window. In part 2 of Figure 2, a 3×3 filter slides over the digit one stop at a time; at each stop it writes one number, and the numbers form a new grid called a **feature map** (G-766).

A layer that slides small filters over an image in this way is a **convolutional layer** (G-480); the convolution operation shows the sliding step by step. A **convolutional neural network (CNN)** (G-484) is a network with at least one convolutional layer. CNNs power almost every image and video application: self-driving cars, or scanning chest images for cancer. Yann LeCun is known as the father of CNNs.

2.3 Recurrent neural network (RNN) and LSTM

> **Key point:** Information loops back from a hidden layer, so the network remembers earlier steps of a sequence; LSTM is a stronger variant.

A sentence arrives one word at a time, and each word only makes sense with the words before it. A network for such data needs a memory.

MLPs and CNNs have none. They are **feed-forward networks** (G-775): information moves in one direction only, from the first layer to the last. In a **recurrent neural network (RNN)** (G-1647), the output of a hidden layer is fed back into the network, so earlier inputs influence later ones. In part 3 of Figure 2, the words "the film was good" enter one by one; after each word the red loop carries the hidden layer's output back, and the hidden layer ends up holding all four words.

A popular improved variant is the **LSTM** (G-1123) (long short-term memory). RNNs and LSTMs are used heavily in natural language processing: voice assistants such as Google Now, Siri and Cortana, and chatbots.

2.4 Autoencoder

> **Key point:** Input and output layers have the same size, the middle is narrow; the network learns to squeeze data and rebuild it.

To make an image, a file or a video smaller while keeping its quality as far as possible, a network can be trained to squeeze the data and then rebuild it. In part 4 of Figure 2, a digit of 64 pixel values is squeezed into 8 numbers in the middle, and the right half of the network rebuilds 64 pixel values from those 8 numbers. The rebuilt digit is a little blurred but is clearly the same digit.

Such a network is an **autoencoder** (G-231). Its input and output layers have the same number of nodes, and its hidden layers have fewer. The network learns to pass the data through the narrow middle and rebuild it on the other side.

> **Extra:** The narrow middle layer cannot hold a full copy of the input, so the autoencoder learns to keep only the most important features, and the rebuilt output loses some detail (Goodfellow et al. 2016, §14.1). Trained to rebuild clean inputs from noisy copies, an autoencoder removes noise (§14.5). Inputs that rebuild badly can be flagged as unusual: anomaly detection (Sakurada and Yairi 2014).

2.5 Generative adversarial network (GAN)

> **Key point:** A generator makes fakes, a discriminator judges real or fake; competing, the generator learns to create realistic new data.

The last type creates new data instead of labelling existing data. The idea is a contest between a forger and a judge. In part 5 of Figure 2, one network produces a fake image, the other network compares it with real images and gives its verdict, and the verdict goes back to the first network, which adjusts. Round after round the fakes get better, until the judge can no longer tell.

The contest is a **generative adversarial network (GAN)** (G-840), introduced by Ian Goodfellow and colleagues in 2014 (Goodfellow et al. 2014): two networks playing a game against each other.

- The **generator** (G-843) creates new data, for example photos of faces.
- The **discriminator** (G-620) decides whether each photo is real or fake.

As the game goes on, the generator learns to fool the discriminator. Its output is then new data that never existed: faces of people who never lived, music, stories and handwriting.

3. History of deep learning

> **Key point:** Two booms and two winters: the **perceptron** (G-1486) (1958) and its XOR failure (1969), backpropagation (1986) and its scaling failure, then deep belief nets (2006) and the 2012 ImageNet win.

Deep learning (G-568) is a young field, and many of the people who shaped it are still active. Its history reads like the plot of a film, full of ups and downs: two rises and two falls (Figure 3).

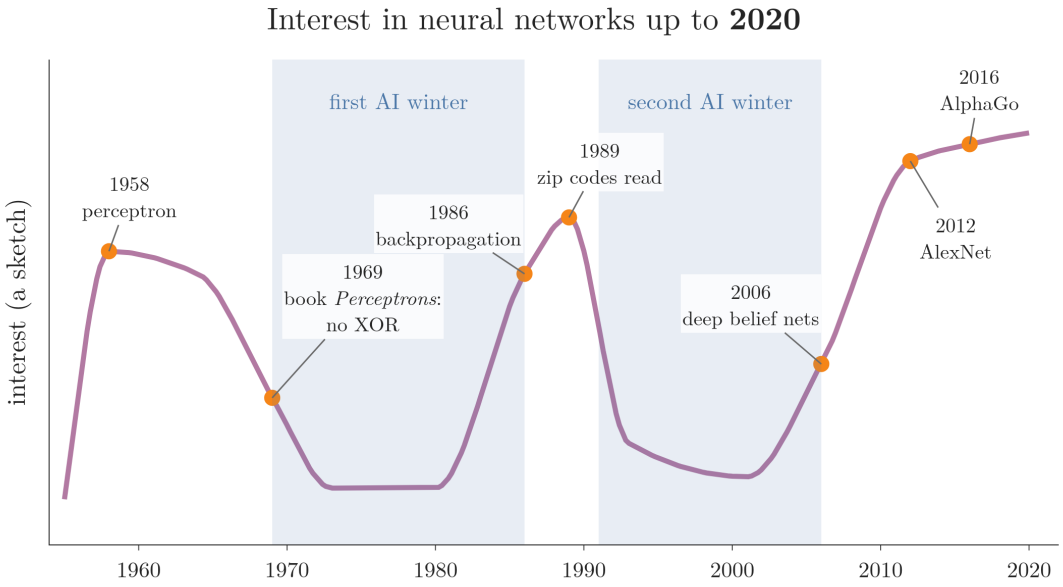


Figure 3: Interest in neural networks drawn year by year, with the two AI winters shaded. The height of the curve is a sketch of the story below, not measured data; the events and years are exact

In Figure 3, watch the curve pause at each orange dot: every rise starts with an invention, and every shaded fall starts when the networks of the time hit a limit. Sections 3.1 to 3.6 follow the curve from left to right.

3.1 The perceptron and the first AI winter

> **Key point:** Rosenblatt's perceptron promised learning machines; Minsky and Papert showed it cannot learn XOR, and funding dried up.

After the Second World War, countries, above all the United States, invested heavily in computer science research. In 1958 the psychologist **Frank Rosenblatt** introduced the **perceptron** (G-1486), a model that learns, loosely based on a neuron in the brain (Rosenblatt 1958). The press hailed it as the true start of artificial intelligence, and Rosenblatt himself made bold claims for it.

In 1969 **Marvin Minsky** and Seymour Papert showed a major limitation: a single perceptron can never learn the **XOR** (G-2134) function, however long it is trained. The reason is that a perceptron separates its two classes with one straight line, and XOR's classes cannot be split by one line (a perceptron trained in code fails on XOR).

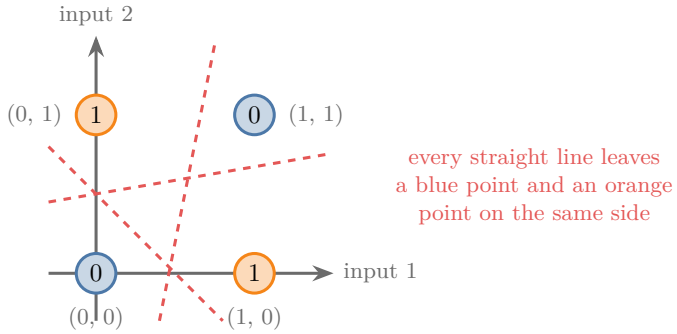


Figure 4: The four XOR points. The output is 1 (orange) when exactly one input is 1. No straight line puts both orange points on one side and both blue points on the other

In Figure 4, try any of the dashed lines: each one leaves a blue point and an orange point on the same side.

Funding, media attention and research on neural networks dropped sharply. A period like this, when interest and money for AI collapse, is called an **AI winter** (G-185); this was the first.

3.2 Backpropagation and the second boom

> **Key point:** Many **perceptrons** (G-1486) in layers can learn non-linear functions, and backpropagation can train them.

In 1986, **Geoffrey Hinton**, now known as a father of **deep learning** (G-568), published with David Rumelhart and Ronald Williams the paper *Learning representations by back-propagating errors* (Rumelhart et al. 1986). The paper's message:

- A single perceptron can only learn linear functions.
- Many perceptrons arranged in layers, with hidden layers, can learn non-linear ones.
- Such a network can be trained with **backpropagation** (G-247): work out how much each weight adds to the error (its derivative) and nudge every weight so the error shrinks (the steps of backpropagation).

Training layered networks with backpropagation overcame the perceptron's biggest flaw, and enthusiasm returned. In 1989 Yann LeCun, a former postdoctoral researcher with Hinton (ACM 2019), used a network trained by backpropagation to read handwritten zip codes from images (LeCun et al. 1989).

Extra: Hinton is the great-great-grandson of the mathematician George Boole, after whom Boolean logic is named. He shared the 2018 Turing Award with Yann LeCun and Yoshua Bengio, and the 2024 Nobel Prize in Physics with John Hopfield (ACM 2019; Nobel Prize 2024).

3.3 Universal approximation

Key point: A network with a hidden layer and enough neurons can approximate any continuous function as closely as we like.

In plain words, a neural network is a machine that bends a curve until the curve fits the data. Each hidden neuron adds one bend, so with enough neurons the curve can follow almost any shape.

The formal statement is the **universal approximation theorem** (G-2052): a neural network with at least one hidden layer and enough neurons can approximate any continuous function, to any accuracy we want. The theorem is why neural networks are called **universal function approximators**.

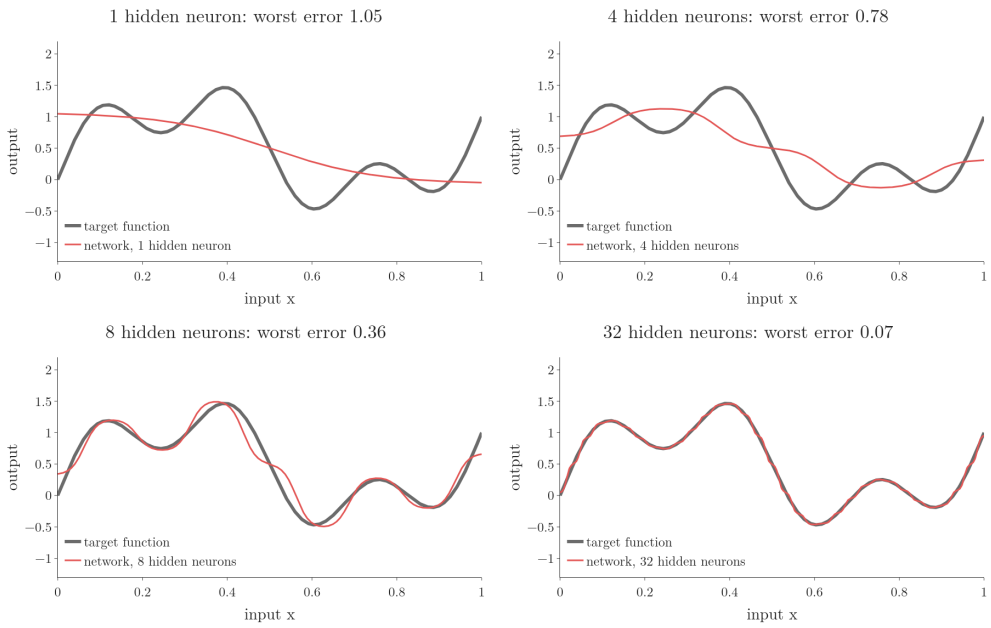


Figure 5: One hidden layer of 1, 2, 4, 8, 16 and 32 sigmoid neurons approximating the same continuous function (grey). The hidden neurons are placed evenly along the input; only the output weights are fitted, by least squares

Figure 5 is a small experiment of our own design: watch the red curve as neurons are added. With one neuron it is a single smooth step; with 32 it follows every bend, and the worst error falls from 1.05 to 0.07.

Extra: The theorem was proved by George Cybenko (1989) and Kurt Hornik (1991). One hidden layer is already enough in principle, though that layer may need an enormous number of neurons; deeper networks often need far fewer. The theorem says a good network exists; it does not promise that training will find it (Goodfellow et al. 2016, §6.4.1).

3.4 The second AI winter

Key point: Networks failed on large problems: too little labelled data, too little computing power, random weight initialisation, and stronger rivals such as SVM and random forests.

The theorem in section 3.3 says a good network exists; it does not say we can train one in practice. By the early 1990s it became clear that networks did not perform well on large problems. Four reasons stood out:

1. **Too little labelled data.**
2. **Too little computing power** to compute all the weights with **backpropagation** (G-247) and add many layers.
3. **No good way to initialise the weights:** they started at random values, which caused errors in training.
4. **Better rivals:** algorithms such as the **support vector machine** (G-1921) and **random forest** (G-1611) trained well on less data.

Interest fell again for about 15 years: the second **AI winter** (G-185).

3.5 Deep belief networks and the name "deep learning"

Key point: In 2006 Hinton showed how to start the weights well, so that networks with many layers could be trained; the field was renamed deep learning.

In 2006, Hinton (with Simon Osindero and Yee-Whye Teh) published a method for training **deep belief networks** (G-567) (Hinton et al. 2006). The method's key idea, **unsupervised pre-training** (G-2059), sets the starting weights with a network trained layer by layer on the inputs alone, without labels (that is what "unsupervised" means), instead of at random. With it, networks with many layers could be trained.

From then on, the field of artificial neural networks was rebranded as **deep learning** (G-568). Between 2006 and 2010, many papers followed.

3.6 2012: ImageNet and after

Key point: In 2012 a deep network trained on GPUs cut the ImageNet error from about 26% to about 15%; from then on **deep learning** (G-568) never looked back.

ImageNet (G-920) is a very large dataset of labelled images, and from 2010 a yearly competition asked teams to classify its images. The best error rates in 2010 and 2011 were

around 28% and 26%.

In 2012, Hinton's team (Alex Krizhevsky, Ilya Sutskever and Hinton) entered a deep network, later called **AlexNet** (G-187), trained on GPUs. Its error was about 15%, against about 26% for the best other entry (Figure 6).

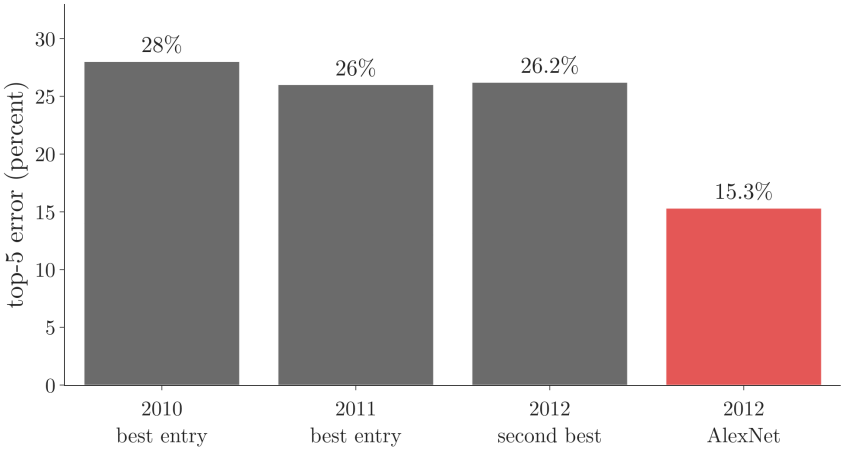


Figure 6: ImageNet top-5 error: the best entries of 2010 and 2011 (about 28% and 26%), and in 2012 the second-best entry against AlexNet

In Figure 6, the first three bars barely move; the red bar is the jump that made companies and researchers take notice. Google, Facebook and other companies took notice, researchers flooded in, and a whole ecosystem grew.

Further milestones:

- **2014:** Ian Goodfellow introduces GANs.
- **2016:** DeepMind's **AlphaGo** beats Lee Sedol, winner of 18 international titles, by 4 games to 1. Go is a board game far more complex than chess: about 250 possible moves per turn against about 35 (Silver et al. 2016).
- **Today:** almost every digital product uses deep learning somewhere, and universities teach it widely.

> **Extra:** The 2012 figure is the top-5 error (the right answer is among the model's five guesses): 15.3% for AlexNet against 26.2% for the second-best entry (Krizhevsky et al. 2012).

The reasons for this final success are the five forces of why deep learning took off:

1. large labelled public datasets;
2. faster hardware (GPUs);
3. easy frameworks;
4. ready-made architectures;
5. a large community.

4. Applications of deep learning

> **Key point:** Since 2012 **deep learning** (G-568) has entered almost every field: driving, games, assistants, translation, photos, and the creative work of GANs.

Deep learning is everywhere: recommendation engines such as Netflix's, chatbots, and much more. Applications of ML covers business uses such as retail and banking; this section lists uses that rely on deep learning.

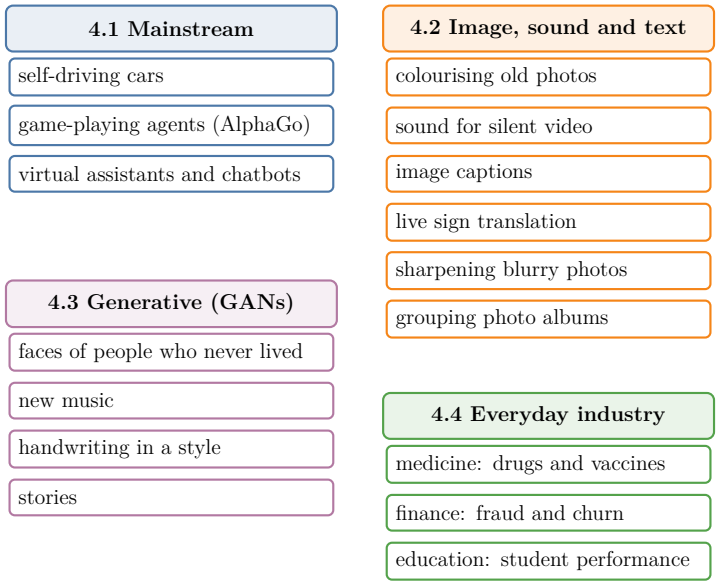


Figure 7: The applications of deep learning in four groups, one per subsection below

Figure 7 is the map of this section: each coloured group is one of the subsections 4.1 to 4.4.

4.1 Mainstream applications

> **Key point:** Self-driving cars, game-playing agents, and virtual assistants.

- **Self-driving cars:** an ordinary car with sensors: cameras on every side, **LiDAR** (G-1083) on the roof to measure the distance to nearby objects, and GPS. A computer runs **deep learning** (G-568) models on all this input and controls the car's mechanics (Figure 8). Waymo (part of Google's parent company Alphabet), Tesla and Uber have worked on such cars.
- **Game-playing agents: deep reinforcement learning** (G-572), the combination of deep learning and **reinforcement learning** (G-1660), raised the skill of game programs enormously. AlphaGo, by Google's DeepMind, beat the Go champion of its time, in a game with more possible positions than there are atoms in the universe.
- **Virtual assistants and chatbots:** Google Now, Siri and Cortana understand speech and follow the context of a conversation. Their quality jumped once deep learning, such

as RNNs, became mainstream around 2015, and most companies now run chatbots built on it.

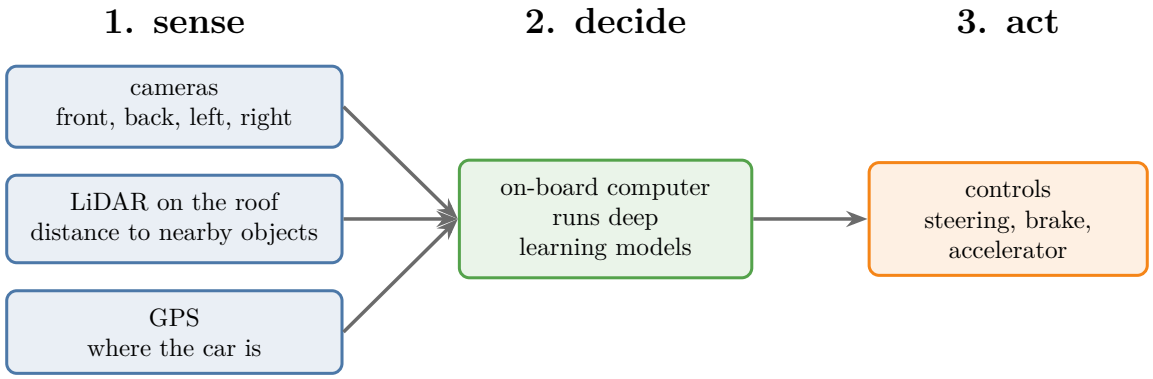


Figure 8: A self-driving car in three steps: the sensors read the surroundings, the computer's deep learning models decide, and the controls act

In Figure 8, follow the arrows from left to right: every sensor feeds the same computer, and only the computer touches the steering, brake and accelerator.

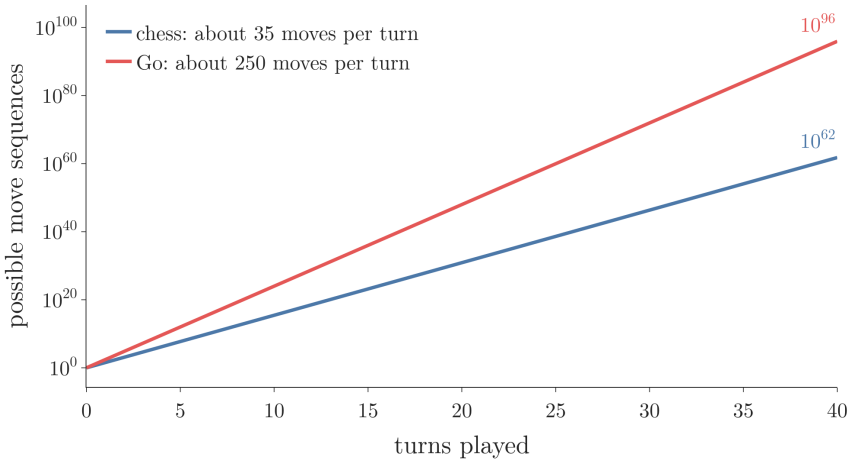


Figure 9: Possible sequences of moves after a number of turns, with about 35 choices per turn (chess) and about 250 (Go), on a log scale

Figure 9 shows why Go was the harder target. Its vertical axis is a log scale: equal distances up stand for equal multiplications (each labelled tick is 10^20 times the one below it), so a straight line means growth by the same factor every turn. After 40 turns, the number of possible move sequences is:

- chess: about 10^{62} (a 1 followed by 62 zeros);
- Go: about 10^{96} (a 1 followed by 96 zeros).

Both are far too many for any program to search one by one, and Go has 10^{34} times more.

4.2 Image, sound and text applications

> **Key point:** Networks colour old photos, add sound to silent video, caption images, translate signs live, sharpen blurry photos and sort photo albums.

- **Image colourisation:** a network predicts the colour of every pixel of a black-and-white photo; used to colour old films.
- **Sound for silent video:** a network watches a silent clip and generates matching sound.
- **Image caption generation:** a network writes a sentence describing what happens in a photo; similar models write taglines from movie posters.
- **Text translation:** point a phone camera at a sign in an unknown language and read it translated in real time. Much of Google Translate runs on **deep learning** (G-568), which helps travellers with signboards and bus numbers.
- **Pixel restoration:** Google Brain researchers turned very low-resolution photos into sharper, higher-resolution ones by predicting the missing pixels from their neighbours, which revives old, blurry photos.
- **Object detection and identification:** Google Photos detects the people, pets and objects in our photos, identifies them, and groups them into albums automatically. Google Photos even assembles a slideshow of a growing child, with a caption, without anyone asking.

4.3 Generative applications

> **Key point:** GANs create new faces, music, handwriting and stories.

GANs generate data that never existed: photos of people who never lived, a prediction of what a couple's baby would look like, new music, handwriting in a given style, and stories (one AI-written story was made into a short film). **DeepDream**, another Google technique (not a GAN), changes an image so that a network's layer responds to it more strongly, which exaggerates the patterns the layer looks for and produces dream-like pictures (Olah et al. 2017).

4.4 Everyday industry uses

> **Key point:** Medicine, finance and education use **deep learning** (G-568) for drugs, fraud and student performance.

- **Medicine:** drug research, genetics and vaccine development.
- **Finance:** fraud detection and customer churn prediction.
- **Education:** measuring and predicting students' performance.

5. Summary

Type	Structure	Good for	Why
MLP	perceptrons in layers, feed-forward	supervised learning on tables; non-linear relationships	hidden layers capture non-linear relationships a single perceptron cannot
CNN	at least one convolutional layer	images, video	what matters in an image is small and can sit anywhere, so a small filter slides over it
RNN, LSTM	output fed back into the network	text, speech, sequences	each word makes sense only with the words before it, so the network needs a memory
Autoencoder	narrow middle layer	compression, denoising	the narrow middle can keep only the most important features
GAN	generator against discriminator	generating new images, music, text	the contest pushes the generator until the judge cannot tell its fakes from real data

Year	Event	Why it mattered
1958	Rosenblatt's perceptron	a model that learns, hailed as the start of AI
1969	Minsky and Papert: a perceptron cannot learn XOR; first AI winter	a perceptron splits classes with one straight line, and XOR's classes need more
1986	Rumelhart, Hinton and Williams: backpropagation trains multi-layer networks	layered networks learn non-linear functions, which overcame the perceptron's biggest flaw
1989	LeCun reads handwritten zip codes	a network trained by backpropagation solved a real task

Year	Event	Why it mattered
1990s	second AI winter: too little data and computing power, random initialisation, SVM and random forests	networks failed on large problems while rivals trained well on less data
2006	Hinton: deep belief networks and pre-training; the name "deep learning"	good starting weights let networks with many layers train
2012	AlexNet on GPUs nearly halves the ImageNet error	companies and researchers took notice, and the field never looked back
2016	AlphaGo beats Lee Sedol	Go is far more complex than chess

- **Feed-forward networks** (G-775) (MLP, CNN) pass information one way; RNNs loop it back, so only RNNs remember the earlier steps of a sequence.
- A network with a hidden layer and enough neurons can approximate any continuous function, which is why networks are called universal function approximators; the theorem says a good network exists, not that training will find it.
- Data, hardware, frameworks, ready-made architectures and community ended the winters, because the second winter came from too little labelled data, too little computing power and poor starting weights.
- So each type of network suits one kind of data or task, and networks rose and fell twice before data, GPUs and better training made **deep learning** (G-568) take off in 2012.

6. Sources

Built from

- CampusX, "Types of Neural Networks | History of Deep Learning | Applications of Deep Learning", YouTube, https://www.youtube.com/watch?v=fne_UE7hDn0
- StatQuest with Josh Starmer, "The Essential Main Ideas of Neural Networks", YouTube, https://www.youtube.com/watch?v=CqOfi41LfDw (the plain-words view in section 3.3: a network fits a curve to the data)

Other references

- Goodfellow, Bengio and Courville, *Deep Learning*, MIT Press, 2016: §6.4.1 (universal approximation, depth), §14.1 and §14.5 (autoencoders, denoising).
- Goodfellow et al., "Generative Adversarial Nets", NeurIPS 2014.
- Sakurada and Yairi, "Anomaly Detection Using Autoencoders with Nonlinear Dimensionality Reduction", MLSDA workshop, 2014.
- Rosenblatt, "The perceptron: a probabilistic model for information storage and organization in the brain", *Psychological Review*, 1958.
- Minsky and Papert, *Perceptrons*, MIT Press, 1969.

- Rumelhart, Hinton and Williams, "Learning representations by back-propagating errors", *Nature*, 1986.
- Hinton, Osindero and Teh, "A Fast Learning Algorithm for Deep Belief Nets", *Neural Computation*, 2006.
- LeCun et al., "Backpropagation Applied to Handwritten Zip Code Recognition", *Neural Computation*, 1989.
- ACM, 2018 A.M. Turing Award announcement, March 2019, and laureate profile of Yann LeCun (postdoctoral work with Hinton in Toronto, 1987 to 1988), amturing.acm.org; The Nobel Prize in Physics 2024, nobelprize.org.
- Cybenko, "Approximation by Superpositions of a Sigmoidal Function", 1989; Hornik, "Approximation Capabilities of Multilayer Feedforward Networks", 1991.
- Krizhevsky, Sutskever and Hinton, "ImageNet Classification with Deep Convolutional Neural Networks", NeurIPS 2012.
- Silver et al., "Mastering the game of Go with deep neural networks and tree search", *Nature*, 2016 (search-tree sizes of Go and chess).
- Olah, Mordvintsev and Schubert, "Feature Visualization", *Distill*, 2017 (describes DeepDream, introduced by Mordvintsev, Olah and Tyka in 2015).

7. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Autoencoder (G-231)	A network with a narrow middle layer that learns to compress data and rebuild it.
Multi-layer perceptron (MLP) (G-1270)	Many perceptrons organised in layers: input, hidden and output.
Feed-forward network (G-775)	A network in which information moves only from the first layer to the last.
Recurrent neural network (RNN) (G-1647)	A network whose hidden-layer output is fed back in, so it remembers earlier steps of a sequence.
Generative adversarial network (GAN) (G-840)	A generator and a discriminator competing, so that the generator learns to create realistic new data.
Generator (G-843)	The GAN network that creates new data.
Discriminator (G-620)	In a GAN, the network that judges whether data is real or fake.
XOR (G-2134)	The logic function that outputs 1 when exactly one of two inputs is 1.
AI winter (G-185)	A period when funding and interest in AI collapse.

Term	Meaning
Universal approximation theorem (G-2052)	A network with a hidden layer and enough neurons can approximate any continuous function.
Deep belief network (G-567)	A many-layered network from Hinton and colleagues (2006) whose starting weights were set by unsupervised pre-training, one layer at a time; its success led the field to be renamed deep learning.
Unsupervised pre-training (G-2059)	Setting a network's starting weights with a network trained layer by layer, instead of at random.
ImageNet (G-920)	A very large labelled image dataset with a yearly classification competition.
LiDAR (G-1083)	A sensor that measures distances to nearby objects with laser light.
Deep reinforcement learning (G-572)	Reinforcement learning (an agent learning from rewards by trial and error) that uses deep neural networks; it raised the skill of game-playing programs.
Perceptron (G-1486)	The smallest building block of a neural network, one artificial neuron: it multiplies each input by a weight, adds the results and turns the sum into an output.
Deep Learning (DL) (G-568)	Machine Learning that uses neural networks with many layers; finds features by itself.
Long short-term memory (LSTM) (G-1123)	A recurrent network (RNN) that carries two memories from one time step to the next, a long-term one (the cell state) and a short-term one (the hidden state), with gates that control what each memory keeps and passes on.
Feature map (CNN) (G-766)	The grid of numbers a filter produces; large where the filter's pattern is present.
Convolution layer (G-480)	A layer that slides small filters over its input to find features.
Convolutional neural network (CNN) (G-484)	A neural network that slides small filters over its input to find patterns such as edges, using at least one convolutional layer; the standard network for images.

Term	Meaning
Backpropagation (G-247)	The algorithm that computes the gradient of the loss with respect to every weight and bias by applying the chain rule backward from the output; gradient descent then uses these gradients to train the network.
Support vector machine (SVM) (G-1921)	A classifier that separates the classes with the line, or flat surface in more dimensions (a hyperplane), that leaves the widest gap (margin) between the classes.
Random forest (G-1611)	Bagging with decision trees as the base models: many trees, each trained on a random sample of the rows with a random choice of features at every split, vote (classification) or are averaged (regression), which lowers the variance of a single tree.
AlexNet (G-187)	A deep convolutional neural network, trained on GPUs, that won the ImageNet contest in 2012 with about 15% error against about 26% for the next best entry, starting the current wave of deep learning.
Reinforcement learning (G-1660)	Learning by acting and receiving rewards or punishments.